Isolating the Effect of Energy Aware Routing on Network Lifetime in Mobile Ad Hoc Networks

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Austin Rimmer & Bryce Desabrais

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1. Introduction

Mobile Ad Hoc Networks (MANETs) are decentralized wireless networks in which mobile devices communicate directly with one another without any fixed infrastructure. Each node acts as both a host and a router, making routing decisions in a distributed manner. MANETs are well suited to scenarios where infrastructure has been destroyed or was never built, such as disaster relief, military coordination, and search and rescue operations.

Because MANET nodes are battery powered and typically cannot recharge during operation, energy management is one of the most critical design concerns. The standard AODV routing protocol selects paths based purely on hop count, ignoring the residual energy of intermediate nodes entirely. This means heavily used central nodes drain their batteries far faster than peripheral ones, causing network partitions long before the network's total energy budget is exhausted.

Recent research has proposed various solutions to this problem, ranging from simple energy aware route metrics [1] to complex frameworks involving trust mechanisms, swarm intelligence [2], and graph neural networks [3]. However, many of these advanced systems bundle energy awareness with other components like trust evaluation and AI based optimization, making it difficult to determine how much of the performance improvement comes from energy awareness alone.

This project addresses that gap by isolating the energy aware routing component and evaluating it deeply. We implement and compare two AODV style routing algorithms in a custom Python based MANET simulator: a hop count baseline that mirrors standard AODV, and an energy aware variant inspired by Ket and Hippargi's AODVM formulation [1]. We evaluate both algorithms across 100 paired simulation trials with full statistical significance testing, measuring six metrics, including packet delivery ratio (PDR) and energy variance, that were not measured in the original study.

This report describes the background research that motivated the project, details our simulation design and methodology, presents our results with statistical analysis, and discusses how our findings contribute to the understanding of energy aware routing in MANETs.

2. Background

2.1 Mobile Ad Hoc Networks and AODV Routing

A MANET is represented as a graph where the vertices are mobile devices (nodes) and the edges are wireless communication links between nodes within transmission range of each other. Because all nodes are mobile, the network topology changes constantly, and routing decisions must be made in a distributed manner with no central authority.

The Ad Hoc On-Demand Distance Vector (AODV) protocol is a reactive routing protocol designed specifically for MANETs. When a source node needs to communicate with a destination, it broadcasts a Route Request (RREQ) packet that floods through the network. When the RREQ reaches the destination (or a node with a known route), a Route Reply (RREP) is sent back along the reverse path. AODV selects routes based on hop count, choosing the path with the fewest intermediate nodes. This approach is simple and effective for finding short paths, but it completely ignores the energy state of the nodes along those paths.

The problem that arises from this is that nodes positioned at central locations in the network topology tend to appear on many shortest paths simultaneously. These nodes are used as relays for multiple traffic flows, draining their batteries far faster than peripheral nodes that carry little or no relay traffic. When these central nodes die, the network partitions, and entire regions lose connectivity, even though the majority of nodes still have significant energy remaining. This uneven energy distribution is the core problem that energy aware routing aims to address.

2.2 Paper 1: Modified AODV Energy Aware Routing (Ket & Hippargi, 2016)

Ket and Hippargi [1] directly address the energy imbalance problem in AODV by proposing two modified protocols. Their first protocol, AODVEA (AODV Energy Aware), introduces an energy threshold for forwarding decisions: a node only participates in routing if its residual energy exceeds a set percentage. Route selection uses a max-min energy algorithm, where the node with the minimum remaining energy is identified for each candidate route, and then the route whose minimum energy node has the highest value is chosen.

Their second protocol, AODVM (AODV Modified), uses the same energy threshold but changes the route selection metric. Rather than selecting purely on max-min energy, AODVM selects routes by maximizing the ratio of minimum residual energy to hop count. This combined metric balances energy conservation against path efficiency, preventing the algorithm from selecting unnecessarily long routes just because they pass through high energy nodes.

Simulations were conducted using the NS-2.35 simulator with 50 nodes in a 700x700 meter area using Constant Bit Rate (CBR) traffic at 2 packets per second. They tested across several scenarios varying initial energy (0.3 J, 0.5 J, 0.7 J) and energy threshold percentage (12%, 15%, 20%). Their results showed that AODVM achieved the highest throughput across nearly all scenarios, outperforming both standard AODV and AODVEA. Network lifetime under AODVM was better than standard AODV but slightly lower than AODVEA, since AODVEA is more aggressive about conserving energy at the cost of fewer available routes.

However, the study has some limitations that our project directly addresses. First, PDR and per node energy consumption were not measured. PDR is arguably the most fundamental quality of service metric in any routing study, and energy variance across nodes is the most direct indicator of whether load is being distributed fairly. Second, each scenario was run only once, with no repeated trials or statistical analysis. This means their results may be sensitive to the specific random seed used for node placement and mobility, and it is difficult to determine from their paper whether the observed differences are consistent or incidental.

2.3 Paper 2: Energy-Aware and Trust-Based Secure Routing Using Swarm Intelligence (Hemalatha et al., 2026)

Hemalatha et al. [2] propose an integrated framework combining energy awareness, trust management, and swarm intelligence based optimization for MANETs. Their system consists of four modules: Communication Behavioral Impact Rate (CBIR) for analyzing node behavior, Energy Aware Macro Channel Selection (EAMCS) for energy efficient channel allocation, Optimal Swarm Intelligence Based Route Selection (OSIBRS) for finding routing paths using ant colony and particle swarm optimization, and Cross Mutation Self Scheduling Routing Protocol (CMSSRP) for congestion management and energy efficient routing.

Their experimental evaluation, conducted in NS-2 with up to 50 nodes, demonstrated a 95.7% improvement in network lifetime compared to conventional methods, along with improvements in throughput and reduced delay. These results are impressive, but the tightly integrated nature of their framework means it is difficult to attribute the improvements to any single component. The energy awareness, trust mechanisms, and swarm intelligence all operate together, so the contribution of energy aware routing alone cannot be isolated from the trust and AI components.

2.4 Paper 3: Trust-Aware Energy-Efficient Routing Using Graph Neural Networks (Maya et al., 2025)

Maya et al. [3] propose EER-MANET-EFIAGNN, a framework that integrates Fast Continual Multi-View Clustering (FCMVC) for trust driven cluster head selection, an Explicit Feature Interaction Graph Neural Network (EFIAGNN) for intelligent routing decisions, and a Horned Lizard Optimization Algorithm (HLOA) for parameter tuning. Their simulation in NS-2.35 with 100 nodes demonstrated improvements in energy efficiency, delay, throughput, and detection rate over baseline methods.

As with the swarm intelligence approach, the GNN based framework bundles energy awareness inseparably with trust evaluation and machine learning optimization. Additionally, the computational complexity of GNN inference and swarm intelligence may be impractical for resource constrained MANET nodes, particularly in emergency deployments. This leads to the question that our project directly addresses, how much of the performance improvement reported by these sophisticated systems comes from the energy awareness component alone, and can a simpler formulation achieve meaningful gains without the overhead?

3. Problem and Solution

3.1 Problem Statement

Ket and Hippargi [1] demonstrate that combining hop count and residual energy into a single routing metric improves throughput and network lifetime compared to standard AODV. However, their evaluation leaves important gaps: PDR and energy variance were not measured, only a single simulation run was performed per scenario with no statistical analysis, and the results may not generalize across different random network configurations.

Meanwhile, more recent works [2][3] confirm that energy aware routing contributes to performance gains, but in both cases energy awareness is inseparable from trust and AI components. Our project addresses these gaps by reimplementing the core comparison from Ket and Hippargi, specifically hop count routing versus a combined energy and hop count routing metric, and evaluating it using 100 paired trials, statistical significance testing, and a broader set of metrics.

Our research question is, can a simplified energy aware modification to AODV style routing significantly improve network lifetime and energy fairness compared to standard hop count routing, and how robust are those improvements across randomized simulation conditions?

3.2 Routing Algorithms

We implement two routing algorithms that operate on the same network graph, mobility model, energy model, and traffic generation process. The only variable between experimental conditions is the routing decision logic.

Algorithm 1 - Hop Count Baseline (AODV): Given a source-destination pair, this algorithm will select the minimum hop path using Dijkstra's algorithm. All edges have equal weight, so the result is the path with the fewest intermediate nodes. This mirrors the route selection behaviour of standard AODV and performs as our control condition. In our implementation, we use the NetworkX library's shortest path function, which computes the same result that AODV's flooding based route discovery would produce, without simulating the protocol overhead of RREQ/RREP packets.

Algorithm 2 - Energy Aware Routing (AODVM-style): This algorithm selects paths using a weighted cost function applied via Dijkstra's algorithm. For each directed edge from node u to node v, the cost is defined as:

$$\mathbf{cost(u \rightarrow v) = \alpha + \beta / energy(v)}$$

where α is the hop count weight, β is the inverse energy penalty weight, and energy(v) is the residual energy of the receiving node. When Dijkstra minimizes the total path cost, it produces:

$$\mathbf{total\_cost = \alpha \times hops + \beta \times \Sigma(1/energy(each\ node\ on\ path))}$$

This formulation increases the cost of routing through nodes with low energy, which steers traffic toward nodes with higher residual energy. The α term ensures that longer paths are penalized, preventing extreme detours. This matches the intuition of Ket and Hippargi's AODVM, which also combines hop count and residual energy into a single metric, though the specific mathematical form differs. Their AODVM maximizes min_energy/hop_count along the path, while our formulation minimizes the sum of per edge costs. Both approaches combine the same two signals and achieve the same goal of balancing energy conservation against path efficiency.

The energy aware routing function builds a directed weighted graph because the cost of traversing an edge depends on direction. For example, going from node A to node B costs based on B's energy (since B is the receiver), while going from B to A costs based on A's energy. This asymmetry is important for correctly modelling the energy impact of routing decisions.

3.3 Simulation Design

Our simulation models a MANET of 30 mobile nodes operating in a 500 × 500 meter bounded area. Nodes are initialized at random positions with initial energy levels drawn from a uniform distribution U(80, 100). Two nodes are neighbours if their Euclidean distance falls below the transmission range of 200 meters. The network graph is recomputed at every timestep to reflect node movement.

Nodes follow the Random Waypoint mobility model. At each timestep, every living node moves toward its current waypoint at a speed of 2 meters per step. Upon arrival, a new random waypoint is selected. This model is the standard MANET benchmark and is the same class of mobility used in Ket and Hippargi's simulation [1], enabling a degree of comparability with their results.

The energy model assigns the following perpacket costs: transmission costs 0.3 units from the sending node, reception costs 0.1 units at the destination, and forwarding (receive then retransmit) costs 0.3 units at each relay node. When a node's energy reaches zero, it is marked as dead and removed from the graph. A warm up period of 30 timesteps allows the network topology to stabilize before energy deduction begins. This mirrors the route establishment phase in real AODV networks where nodes discover each other before heavy traffic begins.

Five source-destination traffic flows are randomly generated at the start of each episode, with each node appearing in at most one flow as either a source or destination. Each flow generates 1 packet per timestep. When a flow's source or destination node dies, the flow is reassigned to two new living nodes, maintaining constant traffic pressure throughout the simulation. This models the continuous generation of new communication sessions in a real network and prevents the simulation from becoming idle as nodes die. Each episode runs for 1000 timesteps.

3.4 Experimental Design

Each algorithm is evaluated over 100 independent simulation trials. The same 100 random seeds (0 through 99) are used for both algorithms, so each pair of runs faces identical initial conditions: the same node positions, the same initial energies, the same traffic flows, and the same mobility patterns. This paired design isolates the effect of the routing strategy from randomness in the simulation environment, which is a methodological improvement over the single run per scenario approach of Ket and Hippargi [1].

All simulation parameters are configurable via command line arguments, enabling researchers to test different network conditions without modifying the code. Parameters include the number of nodes, maximum timesteps, alpha and beta weights, transmission range, number of flows, and node speed. Default values are documented in the project README, and the sensitivity analysis described in Section 4.3 explores the effect of varying alpha and beta.

3.5 Metrics Collected

For each algorithm and simulation trial, we record six metrics:

Metric	Definition
Time to First Node Death	Timestep at which the first node exhausts its energy
Time to 30% Node Loss	Timestep when 30% or more of the nodes have died
Time to Disconnection	Timestep when no traffic flow has a viable path

Packet Delivery Ratio (PDR)	Fraction of generated packets successfully delivered
Average Path Length	Mean number of hops per delivered packet
Energy Variance	Variance of residual energy across living nodes

The first three metrics capture network lifetime at different stages of degradation, early (first death), mid (30% loss), and late (disconnection). PDR measures quality of service, average path length quantifies the routing efficiency tradeoff, and energy variance measures fairness in energy distribution. PDR and energy variance were specifically identified by Ket and Hippargi as future work in their paper [1].

3.6 Statistical Methodology

For each metric, the statistical analysis proceeds as follows. First, the paired differences between the energy aware and hop count values are computed across all 100 trials. The Shapiro-Wilk test checks whether these differences are normally distributed. If the differences are normally distributed ($p > 0.05$), a paired t-test is used; otherwise, the Wilcoxon signed-rank test is used as the alternative, as it does not assume any specific distribution. The significance threshold is set at a standard $p < 0.05$. Cohen's d is reported as an effect size measure, computed as the mean of the paired differences divided by their standard deviation. Typically, d values around 0.2 indicate a small effect, 0.5 a medium effect, and 0.8 or greater a large effect.

3.7 Implementation Details and Challenges

The simulation is implemented in Python using NumPy for numerical operations, NetworkX for graph representation and shortest path computation, Pygame for live interactive visualization, Matplotlib for generating plots, and SciPy for statistical tests. The environment is structured as a Gymnasium compatible reinforcement learning environment, enabling potential future extension with RL based routing policies.

Several challenges were encountered during development. Initially, the simulation exhibited a pattern where a small number of nodes would die in the first few timesteps, followed by no further node deaths for the remainder of the episode. This behavior was primarily caused by the simulation stopping updates after the network became disconnected, which prevented any further energy consumption or node failures. In addition, two design issues contributed to uneven energy usage. First, traffic flows that lost their source or destination node would permanently stop generating packets, reducing the overall traffic load in the network. Second, individual nodes could be assigned as both source and destination across multiple flows, causing them to drain disproportionately fast. These issues were resolved by introducing a warm up period and delaying metric collection until after initial instability, implementing flow replacement (where failed flows are reassigned to active nodes), and enforcing unique flow endpoints so that each node participates in at most one flow. Together, these changes ensured a more consistent traffic load and more realistic energy consumption throughout the simulation.

We also implemented per timestep energy logging and per-packet route selection logging. At every timestep, the energy of all 30 nodes is recorded. For every successfully routed packet, the algorithm used, source, destination, chosen path, and the energy of each node on that path are logged. These logs are saved to JSON files for a sample trial, enabling detailed inspection of the algorithm's decision making process.

4. Analysis

4.1 Primary Results

Table 1 presents the statistical comparison of all six metrics across 100 paired trials.

Metric	Hop Count (mean +- SD)	Energy Aware (mean ± SD)	Test	p-value	Cohen's d	Sig?
Time to First Death	249.76 ± 34.94	299.13 ± 18.94	Wilcoxon	< 0.001	+1.62	Yes
Time to 30% Node Loss	487.60 ± 41.43	551.64 ± 34.73	Paired t-test	< 0.001	+1.48	Yes
Time to Disconnection	777.11 ± 113.63	780.12 ± 71.34	Paired t-test	0.809	+0.02	No
Packet Delivery Ratio	0.779 ± 0.050	0.783 ± 0.051	Wilcoxon	0.839	+0.06	No
Average Path Length	2.026 ± 0.148	2.082 ± 0.140	Paired t-test	< 0.001	+0.41	Yes
Energy Variance	254.37 ± 178.60	91.10 ± 120.46	Paired t-test	< 0.001	-0.77	Yes

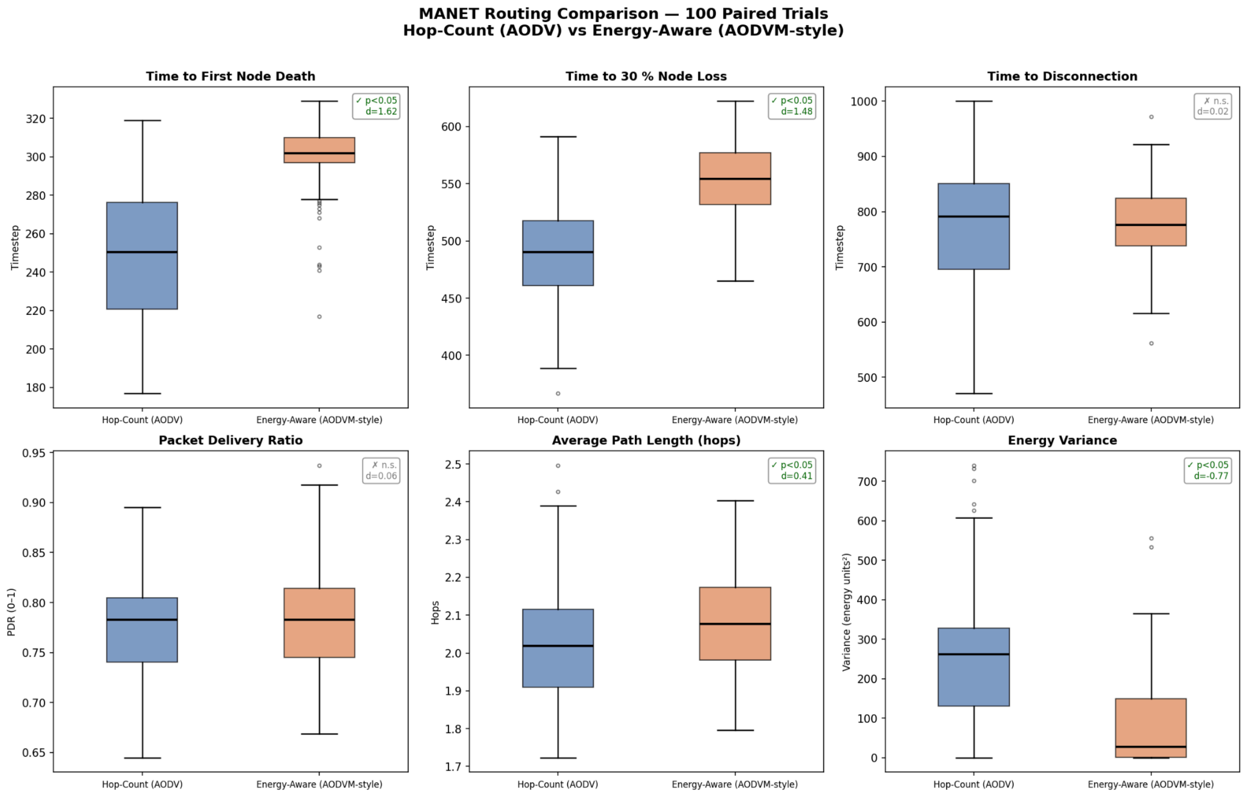

Figure 1: Comparison of Six Performance Metrics Across 100 Paired Trials

Four of the six metrics show statistically significant differences between the two algorithms.

Time to First Node Death: The energy aware algorithm delays the first node death by approximately 50 timesteps on average (250 vs 299), with a very large effect size (d = 1.62). This is our strongest result and directly demonstrates that energy aware routing extends early stage network lifetime by distributing energy consumption more evenly across relay nodes.

Time to 30% Node Loss: The energy aware algorithm delays 30% node loss by approximately 64 timesteps (488 vs 552), with another very large effect size (d = 1.48). This shows the benefit is not limited to the first death but persists into mid stage network degradation. The network maintains a higher node count for a longer period under energy aware routing.

Time to Disconnection: No significant difference (p = 0.809, d = 0.02). Both algorithms disconnect at approximately the same time (~780 steps). This is expected because disconnection is primarily driven by node density relative to the transmission range. Once enough total energy has been consumed and enough nodes have died, the remaining nodes are too spread out to form connected paths regardless of which specific nodes survived. The energy aware algorithm redistributes energy consumption but cannot change the total energy budget of the network.

Packet Delivery Ratio: No significant difference (p = 0.839, d = 0.06). Both algorithms deliver packets at approximately the same rate (~78%). This is a positive finding because it demonstrates that the energy aware modification does not come at the cost of reduced delivery performance. Ket and Hippargi's AODVEA protocol sacrificed throughput for energy conservation [1]; our combined metric avoids this tradeoff, consistent with their AODVM findings.

Average Path Length: The energy aware algorithm produces slightly longer paths (2.08 vs 2.03 hops, d = 0.41). This is the expected tradeoff: by routing around low energy nodes, the algorithm occasionally takes longer paths. However, the difference is small (0.06 hops on average) and does not meaningfully impact delivery performance, as confirmed by the non-significant PDR result.

Energy Variance: The energy aware algorithm reduces energy variance by approximately 64% (254 vs 91), with a large effect size (d = -0.77). This is the most direct measure of the algorithm's intended effect: distributing energy consumption more fairly across all nodes. Lower variance means less disparity between the most drained and least drained nodes, which is exactly what incorporating residual energy into routing decisions should achieve.

4.2 Network Survival and Energy Fairness Over Time

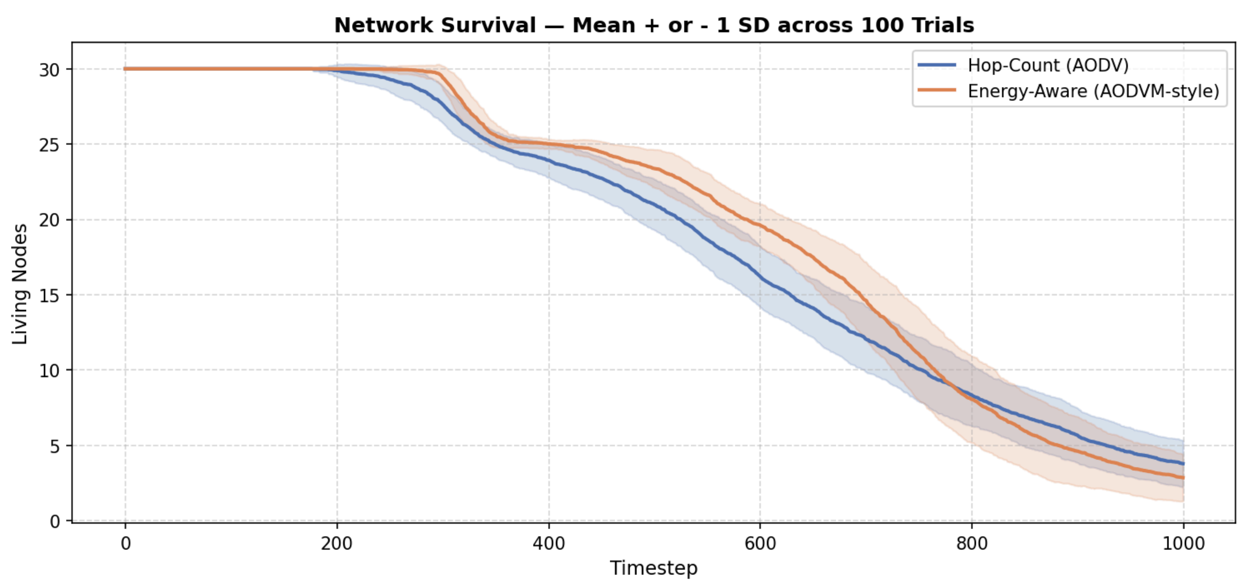

Figure 2: Mean Living Node Count Over Time (±1 std)

The alive over time plot shows the mean number of living nodes across all 100 trials at each timestep, with shaded bands indicating one standard deviation. Both algorithms maintain all 30 nodes through the warmup period (steps 0-30). The hop count algorithm begins losing nodes around step 230, while the energy aware algorithm delays the onset of node deaths to approximately step 280. From step 280 to approximately step 850, the energy aware network consistently maintains 1-3 more living nodes than the hop count network. This gap represents the improved operational quality of the network during its active service life, even though both algorithms converge to similar node counts by the end of the simulation.

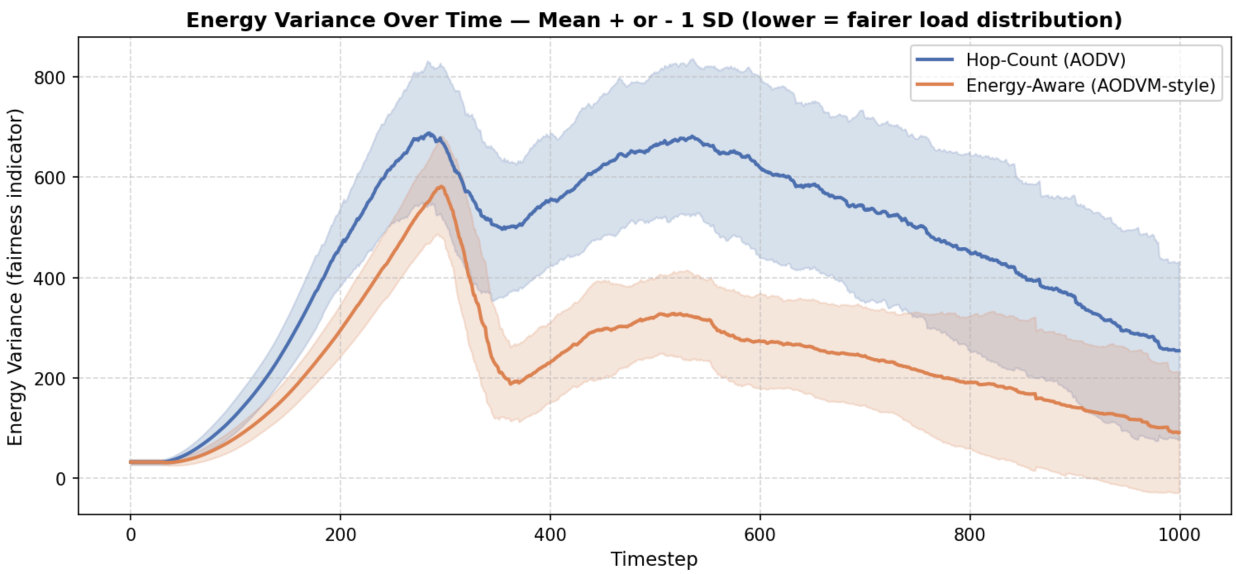

Figure 3: Energy Variance Over Time (±1 std)

The energy variance over time plot reveals an even more dramatic difference. The hop-count algorithm's energy variance peaks at approximately 680 around step 280, then remains elevated between 500-660 through the middle of the simulation before gradually declining as the remaining nodes also drain. The energy aware algorithm's variance peaks at approximately 570 around the same time, but then drops sharply to 200-300 and remains consistently lower throughout the simulation. This sustained separation confirms that energy aware routing is actively distributing load more evenly across the network at every stage of the simulation.

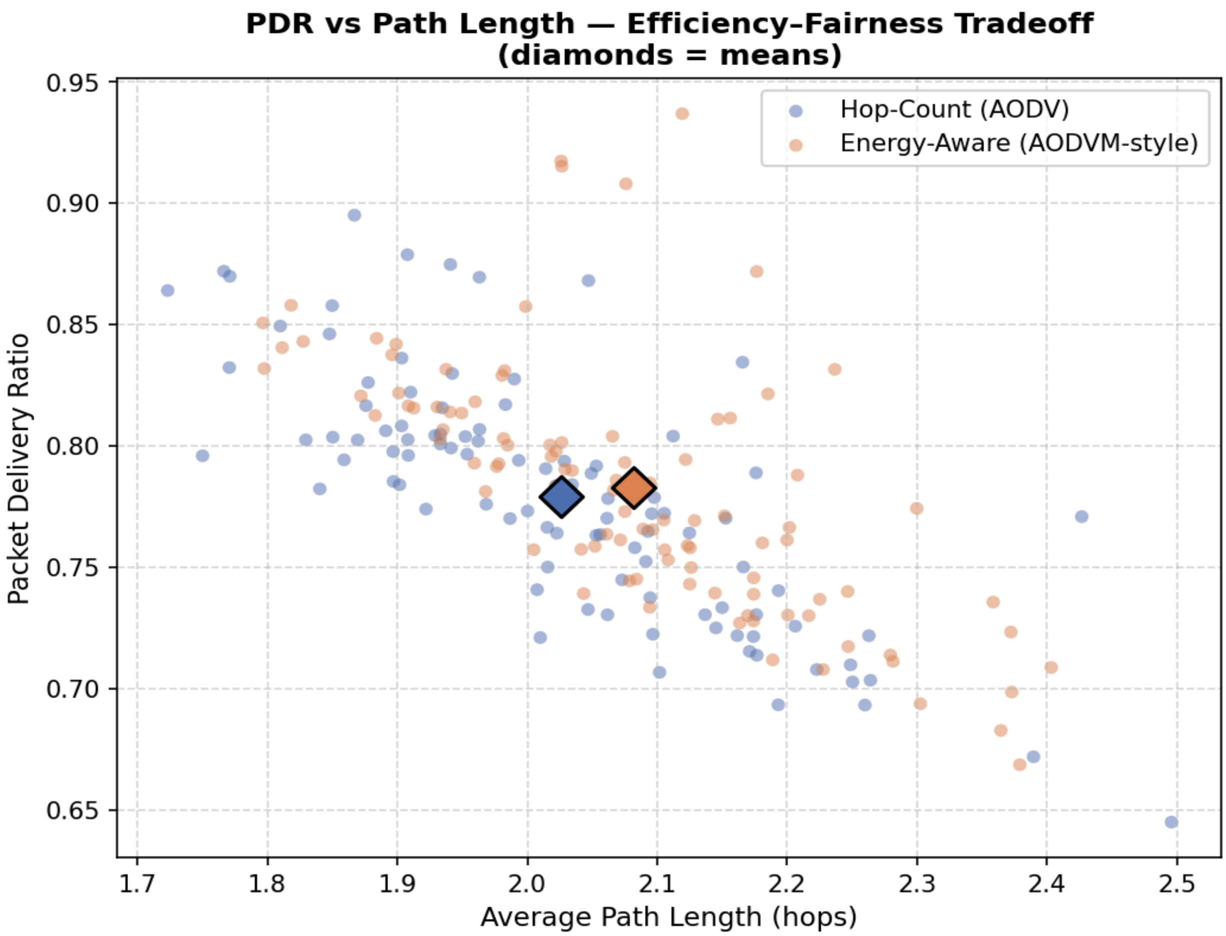

Figure 4: Packet Delivery Ratio vs Average Path Length

The PDR versus path length scatter plot shows the tradeoff between routing efficiency and delivery performance across individual trials. Both algorithms cluster in the same region (1.8-2.4 hops, 0.65-0.95 PDR), confirming that the energy aware modification does not significantly change how the algorithm behaves overall. The mean markers for both algorithms are nearly overlapping, reinforcing the finding that delivery performance is preserved.

4.3 Sensitivity Analysis

To verify that our findings are robust across different parameter choices, we conducted a sensitivity analysis varying α across {0.5, 1.0, 2.0, 4.0} and β across {50, 100, 200, 500, 1000}, running 20 trials per combination. This was proposed in our evaluation plan as an analogous check to Ket and Hippargi's variation of their threshold parameter [1].

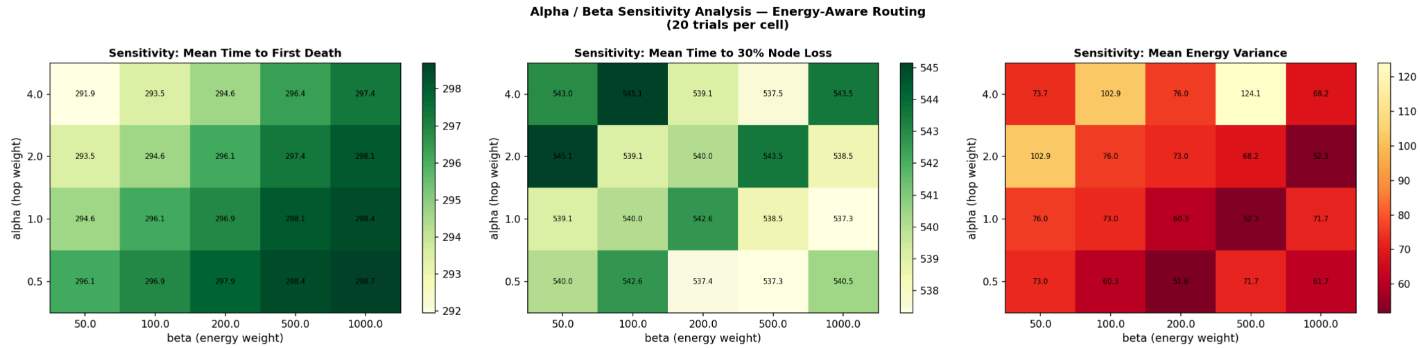

Figure 5: Alpha/Beta Sensitivity Analysis for Energy Aware Routing

The sensitivity heatmaps reveal that the energy aware algorithm improves all three measured metrics (first death time, 30% node loss time, and energy variance) across the entire parameter grid. No combination of α and β produces results worse than the hop count baseline (first death ~250, 30% loss ~488, energy variance ~254). This demonstrates that the benefits of energy aware routing are robust and not an artifact of a particular parameter choice.

The most interesting pattern appears in the energy variance heatmap. Lower α values combined with higher β values produce the lowest variance (best fairness), with the minimum around α = 0.5, β = 200 (variance = 51.6). This makes intuitive sense: a lower α reduces the penalty for longer paths, while a higher β increases the penalty for routing through low energy nodes, together allowing the algorithm to more aggressively distribute load. Conversely, high α with low β approaches hop count behaviour.

The first death time and 30% node loss metrics are relatively stable across the grid, varying by only about 2% and 1.5,% respectively. This suggests that the energy aware algorithm's lifetime benefits are inherent to incorporating energy into the routing metric at all, rather than being highly sensitive to the specific weighting.

In principle, an optimal α/β configuration could be identified by normalizing all three metrics to a common scale and finding the combination that maximizes a weighted composite score. However, the choice of how to weight these metrics depends on the specific deployment scenario. A military operation might prioritize maximum lifetime (favouring higher β), while an emergency communication system might prioritize balanced PDR and path length (favouring moderate β). Our configurable parameter system allows researchers to explore this tradeoff for their specific use case.

4.4 Comparison with Existing Research

Our findings are consistent with Ket and Hippargi's qualitative result [1] that AODVM outperforms standard AODV on network lifetime while maintaining throughput. Our contribution extends their work in several ways: we provide statistical significance testing across 100 paired trials (versus their single run per scenario), we measure PDR and energy variance (which they identified as future work), and we conduct sensitivity analysis across a grid of parameter values.

Our results also support the premise of the more complex frameworks proposed by Hemalatha et al. [2] and Maya et al. [3]: energy-awareness does contribute meaningfully to network lifetime improvements. However, our work demonstrates that a simple additive cost function achieves significant gains ($d = 1.62$ for first death time, $d = -0.77$ for energy variance) without the computational overhead of swarm intelligence, trust mechanisms, or graph neural networks. This suggests that for resource constrained MANET deployments, a lightweight energy aware modification may capture the majority of the benefit without the complexity of more advanced approaches.

5. Conclusion

This project demonstrates that a simple energy aware modification to AODV-style routing produces statistically significant improvements in network lifetime and energy fairness across 100 paired simulation trials. The energy aware algorithm delays the first node death by approximately 50 timesteps ($d = 1.62$), delays 30% node loss by approximately 64 timesteps ($d = 1.48$), and reduces energy variance by 64% ($d = -0.77$), all without any significant cost to packet delivery ratio or disconnection time.

These findings directly extend Ket and Hippargi's work [1] by providing the more thorough statistical evaluation their study lacked, measuring the PDR and energy variance metrics they identified as future work, and confirming that the combined hop count and energy metric produces robust improvements across a range of parameter values. Our results also demonstrate that the energy aware component alone, isolated from the trust and AI mechanisms used in more recent works [2][3], accounts for meaningful performance gains.

The simulation is designed as a configurable research framework. All network parameters (node count, transmission range, energy costs, traffic flows, mobility speed, and the α/β cost function weights) are exposed as command line arguments, enabling researchers to explore different network conditions without modifying the source code. The sensitivity analysis demonstrates how to systematically evaluate parameter choices, and the per-step logging system provides detailed visibility into the algorithm's routing decisions for further study.

Limitations of this work include the abstraction of AODV's route discovery overhead (our simulation computes optimal paths directly without modelling RREQ/RREP flooding), the use of a simplified energy model without MAC-layer effects, and the assumption of symmetric links. Future work could extend this framework with reinforcement learning based routing policies (the Gymnasium compatible environment structure supports this directly), more realistic radio propagation models, or comparison against additional routing protocols.

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